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6882/02

Specimen Paper October/November 2027-2029

2 hours

Additional Material: Calculator

Candidates answer:

Section A: Structured questions on the Question Paper.

Section B: Two Essay questions on the lined space provided.

READ THESE INSTRUCTIONS FIRST

Write your name, centre number and candidate number on all the work you hand in.

Write in **dark blue** or **black** pen.

You may use an **HB** pencil for any diagram(s) or graph(s).

Do not use staples, paper clips, glue or correction fluid.

Answer **all seven** structured questions and attempt **two** essay questions out of the three provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part of the question.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
Total	

Section A: Structured Questions

Answer **all** questions in this section **(70 marks)**

- 1 (a) Outline any **two** reasons why research is important in agriculture development.

1

 2
 [2]

- (b) Over many years different farming systems have evolved; intensive, monoculture and organic farming. One new trend is hydroponics.

- (i) State what is meant by a hydroponic system.

.....
 [1]

- (ii) Suggest **one** advantage of using a hydroponic system.

.....
 [1]

- (c) Describe the contribution of intensive farming systems to the national economy.

.....

 [2]

- (d) Explain how monoculture could result in the use of more fertilisers.

.....

 [2]

(e) Explain how organic farming could be expensive to practice.

.....

.....

.....

..... [2]

[10 Marks]

2 (a) What is meant by agribusiness?

.....
..... [1]

(b) Give any **two** ways farm credit is used in agriculture.

1
2 [2]

(c) State any **two** requirements needed to access a loan.

1
.....
2
..... [2]

(d) Describe resources that are regarded as capital in agricultural production.

.....
.....
.....
..... [2]

(e) Explain the purpose of a farm budget.

.....
.....
.....
..... [2]

[9 marks]

- 3 **Fig. 3.1** shows a farming practice where maize is planted together with beans.



Fig. 3.1

- (a) (i) Name the farming practice shown in **Fig 3.1**.

..... [1]

- (ii) Describe any **two** benefits to the maize crop of the farming practice shown in **Fig 3.1**.

1

.....

2

..... [2]

- (b) Crops grown outside as in **Fig 3.1** involve the use of machinery that can cause soil compaction.

Explain how soil compaction could affect crop production.

.....

.....

.....

..... [2]

- (c) Some crops are grown inside greenhouses (glasshouses) to increase production.

State **two** environmental conditions that are controlled inside a glasshouse.

- 1
.....
- 2
..... [2]

- (d) Crops that are produced for export must comply with certain conditions.

Suggest any **two** standards required when exporting baby vegetables.

- 1
.....
- 2
..... [2]

[9 Marks]

- 4 (a) **Fig. 4.1** shows the response of two maize varieties to increasing levels of NPK fertiliser application.

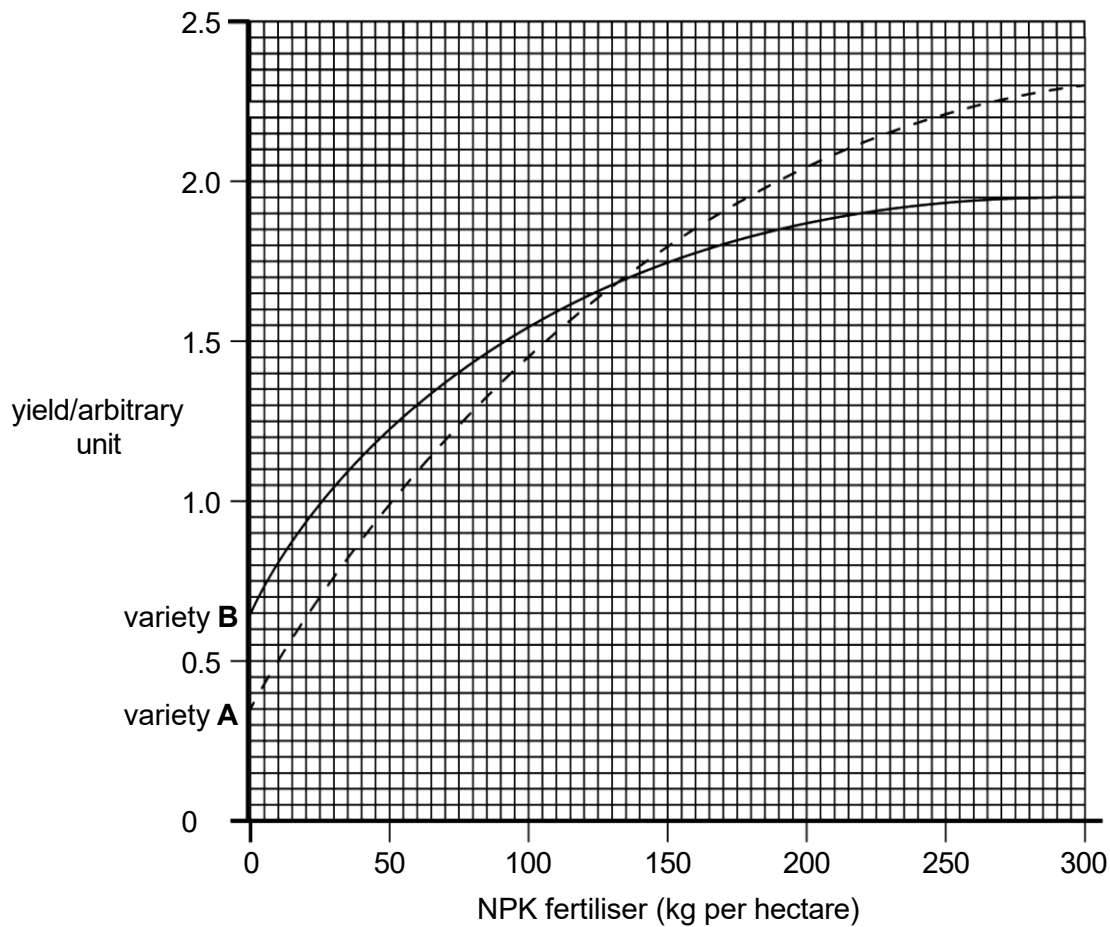


Fig 4.1

- (i) What is the amount of NPK fertiliser applied per hectare to produce a yield of 0.75 units from variety **A** in **Fig. 4.1**?

..... [1]

- (ii) State the amount of NPK fertiliser applied per hectare when the yield from variety **A** and **B** are equal.

..... [1]

- (iii) Calculate the difference in yield between variety **A** and **B** when 190 kg of NPK fertiliser is applied per hectare.

.....

.....

.....

..... [2]

- (b) Before growing a crop and adding fertiliser a soil should be tested.

Fig 4.2 shows the order in which soil samples were taken from a school garden for testing.

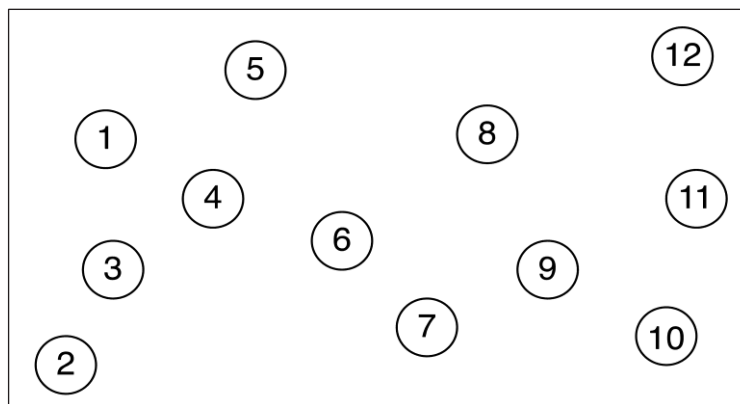


Fig. 4.2

Name the sampling method used in **Fig 4.2**.

..... [1]

- (c) Describe any **two** problems caused by the overuse of inorganic fertilisers to the environment.

1

.....

2

..... [2]

(d) Give any **two** factors that could cause soil to be more acidic.

- 1
.....
- 2
..... [2]

(e) Explain how soil pH affects plant growth.

.....
.....
.....
..... [2]

[11 marks]

5 Fig. 5.1 shows the germination of a seed.

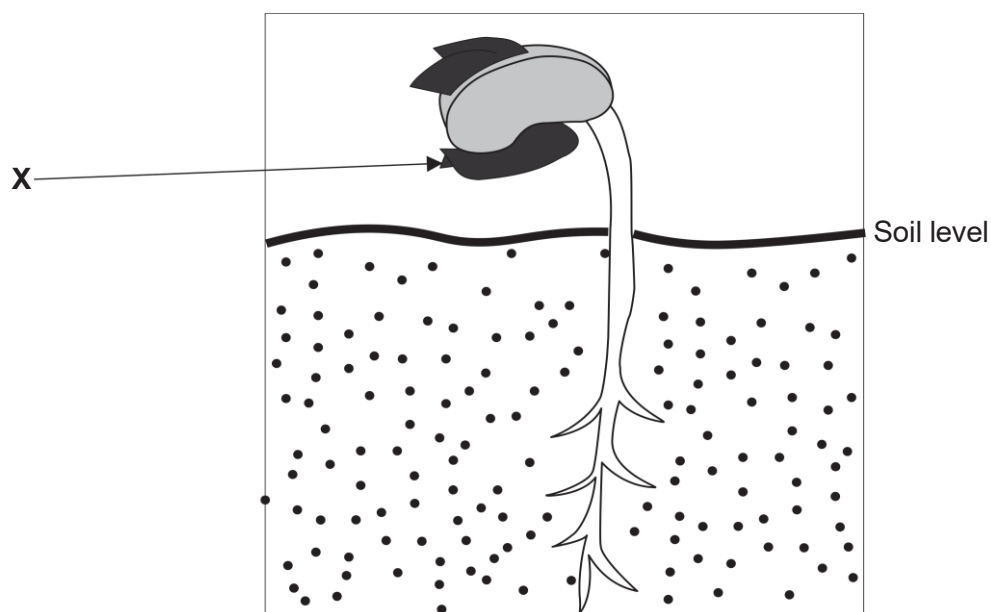


Fig 5.1

(a) (i) Name the part labelled **X** on Fig. 5.1.

.....[1]

(ii) State the **type** of germination shown in Fig. 5.1.

.....[1]

(b) State any **two** conditions necessary for germination.

1

2 [2]

(c) Describe the process by which water moves from the root to the leaves.

.....

..... [3]

(d) Farmers do not always plant crops from seeds.

Suggest any **three** reasons for growing crops using asexual reproduction.

- 1
.....
- 2
.....
- 3
..... [3]

[10 Marks]

6 Fig 6.1 shows an aphid feeding on a plant tissue.

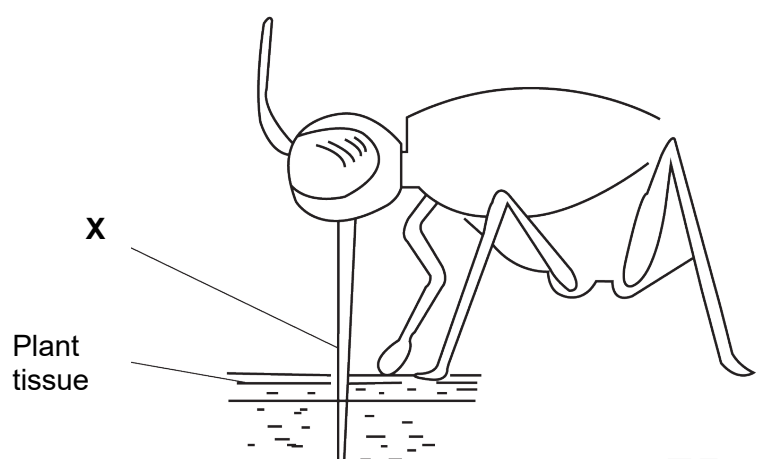


Fig 6.1

(a) Name part **X** in Fig 6.1.

..... [1]

(b) The pest in Fig 6.1 feeds by piercing and sucking.

(i) Suggest **two** ways this reduces crop yields.

1

.....

2

..... [2]

(ii) Explain how this pest is controlled by using a named chemical.

.....

.....

.....

..... [2]

(c) A pesticide application rate on a crop is 1.6 kg in 200 litres of water per hectare.

(i) Calculate how much pesticide is needed for 0.45 hectares? (Show your working)

.....

.....

.....

..... [2]

(ii) Calculate how much of this pesticide should be mixed with 8 litres of water? (Show your working).

.....

.....

.....

..... [2]

(d) Describe bio-intensive integrated pest management.

.....

.....

..... [2]

[11 marks]

7 (a) What is meant by a balanced ration?

.....
..... [1]

(b) Give any **two** functions of proteins in an animal's ration.

1
.....
2
..... [2]

(c) Suggest **two** reasons for adequate water supply in livestock nutrition.

1
.....
2
..... [2]

(d) Fig. 7.1 is a diagram of a digestive system in ruminants.

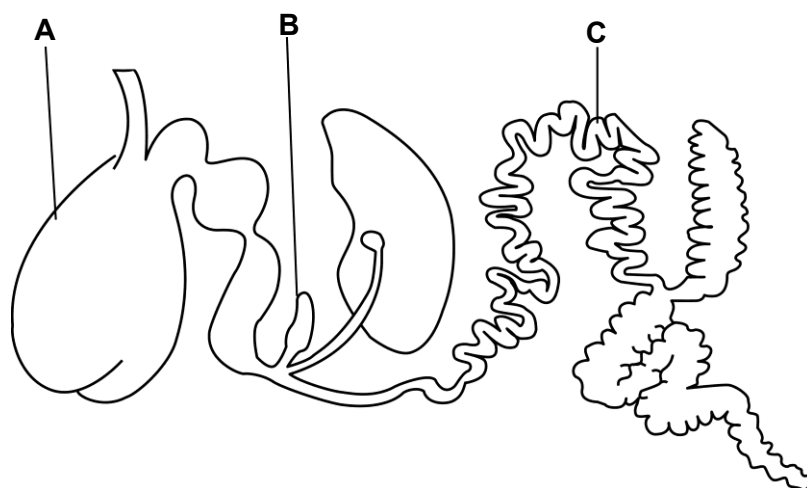


Fig. 7.1

(i) Name organ **B** on Fig. 7.1.

..... [1]

(ii) State the function of the part labelled **C** on Fig. 7.1.

..... [1]

(iii) Suggest any **three** roles of the rumen (**A**) in the process of digestion.

1

.....

2

.....

3

..... [3]

[10 marks]

Section B: Essay Questions

Answer any **two** questions from this section **(30 marks)**

8 (a) Describe any **two** breeding practices in rabbit production. [4]

(b) Explain with details how you would feed a doe at different stages of production using pellets only. [6]

(c) Suggest advantages of using indigenous breeds in livestock production. [5]

[15 marks]

9 (a) Describe the **two** methods that can be used to collect water for agricultural use. [4]

(b) Explain with reasons how water in the soil can be conserved by farmers. [6]

(c) Suggest the positive effects of high temperature on crop growth. [5]

[15 Marks]

10 (a) Describe any **two** factors considered when selecting a suitable site for farm buildings. [4]

(b) Explain with reasons how wooden posts could be preserved. [6]

(c) Suggest advantages of using living fence. [5]

[15 marks]

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